

Tudorica Eric Emanuel

Computer Engineering Student

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[GitHub](#) • [LinkedIn](#) • [Personal portfolio](#)

EDUCATION

BSc in Computer and Information Technology Oct 2025 - Jun 2029 (expected)

University POLITEHNICA of Bucharest (Faculty of Automatic Control and Computers)

- Completed the first year with a **9.61/10 GPA** and received academic distinction awards (top performer) in Operating Systems and Computer Organization & Assembly Language.
- Relevant Coursework:** Data Structures and Algorithms, Computer Programming (C/C++). Additionally completed the CCNA 1 (Introduction to Networks) certification

High School

Sept 2021 - Jun 2025

Colegiul Național "Mihai Viteazul", Bucharest

- Graduated **First in Class** (1st out of 300 graduates) from the Mathematics and Computer Science profile.
- Earned a 9.70/10 average in the Baccalaureate Exam, featuring perfect scores (10/10) in both Mathematics and Computer Science.

RELEVANT EXPERIENCE

Undergraduate Teaching Assistant – Introduction to Operating Systems

Oct 2026 - Present

University POLITEHNICA of Bucharest

- Mentor 1st-year students during hands-on lab sessions, explaining core Linux and operating systems concepts to facilitate their practical learning.
- Co-lead the modernization of course materials by restructuring lab contents and migrating the documentation to a new web platform using Docusaurus.

PROJECTS

Inverted Index Search Engine

March 2026

C

- Engineered a custom file indexing system in C, implementing complex data structures from scratch, including hash maps and tries, for rapid text tokenization and efficient storage.
- Utilized recursive tree traversal techniques to construct an inverted index, heavily optimizing time and space complexity to ensure high-speed search query processing.

Emergency Logistics Management System

May 2026

C

- Developed a logistics management application to handle 112 emergency service calls, dispatch responder units, and dynamically update operational workflows.
- Integrated core Data Structures and Algorithms (DSA) concepts, applying priority queues and optimized routing logic to efficiently process critical, time-sensitive events.

Official Website – Floare de Colț NGO

Spring 2024

AstroJS, CSS

- Designed, built, and deployed a modern, highly responsive web platform (fundatiaecologicafloaredecolt.ro) from scratch to enhance the digital footprint of an ecological foundation.
- Focused on robust front-end architecture and performance optimization, delivering an intuitive user interface to effectively communicate and drive community environmental initiatives.

SKILLS

Programming

C, JavaScript, Bash, Python, HTML, CSS

Tools

Git, Makefile, Markdown, LaTeX, AstroJS

Communication

Romanian (native), English (Cambridge C1)